

TEAM USA

USER MANUAL.



**SMART DIGITAL CLOSET AND
OUTFIT RECOMMENDATION
SYSTEM
VERSION: 2.0**

1. INTRODUCTION

The Smart Digital Closet and Outfit Recommendation System is a console-based application written in C++. It allows users to maintain a digital record of their wardrobe and receive outfit recommendations based on the season, occasion, color compatibility, and appropriate shoe type.

This manual provides a complete, step-by-step guide to help you install, run, and use all features of the system.

Important: This application runs entirely in the console (command prompt / terminal). There is no graphical interface. All interaction is done by typing numbers and text.

2. SYSTEM REQUIREMENTS

Requirement	Details
Operating System	Windows 7/10/11, Linux, or macOS
C++ Compiler	g++ (MinGW on Windows), Clang, or MSVC
IDE (Optional)	Code::Blocks, Dev-C++, Visual Studio, or VS Code
C++ Standard	C++11 or later (use -std=c++11 flag)
Disk Space	Less than 1 MB for the program and data files
RAM	Any modern computer (minimum 256 MB)

3. HOW TO RUN THE PROGRAM

Step 1: Save the Source File

Save the provided C++ code as a file named:

```
smart_closet.cpp
```

Save it in a folder of your choice, for example: C:\MyProjects\SmartCloset\

Step 2: Compile the Program

Using Command Prompt (Windows)

Open Command Prompt. Navigate to the folder where you saved the file:

```
cd C:\MyProjects\SmartCloset
```

Compile the program using g++:

```
g++ -std=c++11 -o smart_closet smart_closet.cpp
```

Using Linux / macOS Terminal

```
g++ -std=c++11 -o smart_closet smart_closet.cpp
```

Using Code::Blocks IDE

1. Open Code::Blocks and create a new Empty Project.
2. Add smart_closet.cpp to the project.
3. Press F9 (Build and Run) to compile and execute.

Step 3: Run the Program

Command Prompt / Terminal

```
./smart_closet (Linux / macOS) smart_closet.exe (Windows)
```

Note: The program will automatically create two data files in the same folder: • users.txt — stores all registered usernames and passwords • clothes.txt — stores all clothing items for all users

4. MAIN MENU

When the program starts, you will see the Main Menu:

```
=====
SMART DIGITAL CLOSET & OUTFIT RECOMMENDATION
=====
1. Login
2. Signup
0. Exit
-----
Choice:
```

Option	Description
1	Log in to an existing account to access your closet.
2	Create a new account (Signup) if you are a first-time user.
0	Exit and close the program.

5. SIGNUP — CREATING A NEW ACCOUNT

If you are using the system for the first time, you must create an account.

How to Signup

1. From the Main Menu, type 2 and press Enter.
2. Enter your desired Username (must be at least 3 characters).
3. Enter your Password (must be at least 4 characters).

4. Re-enter your password to confirm it.
5. If both passwords match, your account will be created successfully.

Sample Signup Session

```
=====                               USER
SIGNUP  =====                               Enter
Username : ahmed
Enter Password : 1234
Confirm Pass : 1234
[SUCCESS] Account created! You can now login.
```

Rules for Signup: • Username must be at least 3 characters long. • Password must be at least 4 characters long. • The two passwords you enter must match exactly. • You cannot use a username that is already taken.

6. LOGIN — ACCESSING YOUR ACCOUNT

If you already have an account, use Login to access your digital closet.

How to Login

1. From the Main Menu, type 1 and press Enter.
2. Enter your registered Username.
3. Enter your Password.
4. If your credentials are correct, you will enter the Closet Menu.

Sample Login Session

```
=====
USER LOGIN =====
Username : ahmed
Password : 1234
[SUCCESS] Welcome, ahmed!
```

If Login Fails: You will see [ERROR] Invalid username or password. Check that you typed your username and password correctly (they are case-sensitive).

7. CLOSET MENU

After a successful login, you will see the Closet Menu:

```
=====
                                DIGITAL CLOSET MENU
                                =====
User: ahmed
1. Add Clothing
2. View My Closet
```

```

3. Edit a Clothing Item
4. Delete a Clothing Item
5. Get Outfit Recommendations
0. Logout ----- Choice:

```

Option	Description
1	Add new clothing items to your wardrobe (Upper, Lower, or Shoes).
2	View all clothing items currently stored in your closet.
3	Get outfit recommendations based on occasion and season.
0	Logout and return to the Main Menu.

8. ADDING CLOTHING TO YOUR CLOSET

Select option 1 from the Closet Menu to add a new clothing item. You will see:

```

=====
ADD CLOTHING TO YOUR CLOSET
=====
1. Add Upper Wear (Shirt, Kameez, Jacket...)
2. Add Lower Wear (Jeans, Shalwar, Trousers...)
3. Add Shoes
0. Back

```

8.1 Adding Upper Wear

Upper Wear includes shirts, t-shirts, kameez, jackets, sweaters, and similar items. You will be asked for the following details:

Field	Example Values	Description
Item Name	White Formal Shirt, Kameez	The name or description of the item
Color	White, Blue, Black, Red	The main color of the item
Gender	Boys / Girls / Unisex	Who the item belongs to
Fabric	Cotton, Wool, Polyester, Silk, Linen	The material of the item
Season	1=Summer, 2=Winter, 3=Both	Enter a number 1, 2, or 3
Occasion	1=Wedding, 2=Sports, 3=Formal, 4=Casual, 5=Party, 6=University	Enter the number for the occasion

8.2 Adding Lower Wear

Lower Wear includes jeans, trousers, shalwar, tights, skirts, and similar items. The fields are the same as Upper Wear. Fabric examples for Lower Wear: Denim, Cotton, Wool, Polyester, Silk.

8.3 Adding Shoes

Shoes have a Shoe Type field instead of Fabric. The following shoe types are supported:

Shoe Type	When to Use
Formal	For Formal, Wedding occasions (Oxford, Loafers, Derby shoes)
Sports	For Sports occasions only (Sneakers, Running shoes)
Casual	For Casual, Party, University occasions
Sandals	For Casual occasions
Heels	For Wedding, Formal, Party occasions (Girls)

Important Fashion Rule: Slippers or Chappals are NOT recognized as valid shoe types in this system. Only the five types listed above are accepted for recommendations.

Sample: Adding a Shoe

```
Item Name      : Oxford Shoes
Color          : Black
Gender         : Boys
Shoe Type      : Formal
Season         : 3
Occasion       : 3
[SUCCESS] Shoes added to your closet!
```

9. VIEWING YOUR CLOSET

Select option 2 from the Closet Menu to view all items in your wardrobe. The system will display each category separately: Upper Wear, Lower Wear, and Shoes.

Each item is displayed using the overridden displayItem() function of its class (this is the Polymorphism feature). Upper items show [UPPER WEAR], lower items show [LOWER WEAR], and shoes show [SHOES].

Sample Output

```
=== UPPER WEAR (2 items) === -----
----- Item #1 [UPPER WEAR]
Name      : White Formal Shirt
Color     : White
Gender    : Boys
Season    : Both (Summer & Winter)
Occasion: Formal
Category: Upper
Fabric    : Cotton
=== SHOES (1 items) ===
```

```
-----  
Item #1    [SHOES]  
Name       : Oxford Shoes  
Color      : Black  
Gender     : Boys  
Season     : Both (Summer & Winter)  
Occasion: Formal  
Category: Shoes  
Shoe Type: Formal
```

If your closet is empty, the system will display: 'Your closet is empty! Add some clothes first.'

10. EDIT YOUR CLOSET

Select option 3 from the Digital Closet Menu to edit clothing items in the wardrobe. The system displays categories including Upper Wear, Lower Wear, and Shoes along with stored items. The user selects an item and updates details such as name, color, gender, fabric/shoe type, season, and occasion. Setter functions are used to update object data, and the updated item is displayed using the overridden displayItem() function, demonstrating Runtime Polymorphism. Finally, the updated data is saved automatically using file handling.

Sample Output

```
=====
                        EDIT CLOTHING ITEM
=====

Select category to edit:
1. Upper Wear  (2 items)
2. Lower Wear  (1 items)
3. Shoes       (1 items)
0. Back

-----

Choice: 1
=== UPPER WEAR ===

-----

[1] Black Shirt | Black | Cotton
-----

[2] White Shirt | White | Cotton
-----

Enter item number to edit (0 to cancel): 1
Editing: Black Shirt
(Press Enter to keep current value)
New Name   [current: Black Shirt]: White Shirt
New Color  [current: Black]: White
New Gender [current: Boys]: Boys
```

```
New Fabric [current: Cotton]: Cotton
Season      (1=Summer 2=Winter 3=Both)   [current: 1]: 1
Occasion    (1=Wedding 2=Sports 3=Formal 4=Casual 5=Party 6=Univ) [current: 3]:
3
[SUCCESS] Item updated successfully!

-----

Updated item:
[UPPER WEAR]
Name       : White Shirt
Color      : White
Gender     : Boys
Season     : Summer
Occasion   : Formal
Category   : Upper
Fabric     : Cotton
```

10. EDIT YOUR CLOSET

Select option 4 from the Digital Closet Menu to remove clothing items from the wardrobe. The system displays all categories and stored items with their item numbers. After selecting an item, the system asks for confirmation before deletion. The selected item is removed from the array and remaining items are shifted to maintain proper indexing. The updated data is then automatically saved into the clothes.txt file using file.

Sample Output

```
=====
          DELETE CLOTHING ITEM
=====

Select category to delete from:
1. Upper Wear   (2 items)
2. Lower Wear   (1 items)
3. Shoes        (1 items)
0. Back

-----

Choice: 1

=== UPPER WEAR ===

-----

[1] White Shirt | White | Cotton
-----

[2] White Shirt | white | Cotton
-----

Enter item number to delete (0 to cancel): 1
```



```
You are about to delete: White Shirt
Are you sure? (1=Yes 0=No): 1
[SUCCESS] Item deleted successfully!
```

10. GETTING OUTFIT RECOMMENDATIONS

This is the main feature of the system. Select option 5 from the Closet Menu to get outfit recommendations.

Before You Start: You must have at least 1 Upper Wear item, 1 Lower Wear item, and 1 pair of Shoes in your closet. If any category is missing, the system will ask you to add items first.

Step 1: Select Occasion

```
Select Occasion for Recommendation: 1 = Wedding 2 = Sports 3 = Formal
4 = Casual 5 = Party 6 = University
Your choice: 3
```

Step 2: Select Season

```
Select Season: 1 = Summer 2 = Winter 3 = Both
Your choice: 1
```

Step 3: View Results

The system searches through all your Upper + Lower + Shoes combinations and shows only the valid ones. An outfit is shown only if ALL of the following conditions are true:

- Occasion Match: All three items (Upper, Lower, Shoes) have the same occasion as selected.
- Season Match: All three items are compatible with the selected season. Items marked 'Both' (3) match any season.
- Color Compatibility: Upper and Lower colors are compatible, AND Lower and Shoes colors are compatible.
- Shoe Validity: The shoe type is appropriate for the selected occasion.

Sample Recommendation Output

```
=====
RECOMMENDED OUTFITS
=====
>> OUTFIT #1 <<
-----
UPPER   : White Formal Shirt (White, Cotton)
LOWER   : Black Trousers (Black, Polyester)
SHOES   : Oxford Shoes (Black, Formal)
SEASON  : Both (Summer & Winter)
FOR     : Formal Occasion
>> OUTFIT #2 <<
-----
```

```

UPPER  : Grey Blazer (Grey, Wool)
LOWER  : Black Trousers (Black, Polyester)
SHOES  : Oxford Shoes (Black, Formal)
SEASON : Both (Summer & Winter)
FOR    : Formal Occasion
=====
Total Outfits Found: 2

```

No Outfits Found?

If no outfits are recommended, the system will suggest:

```

No matching outfits found.
Tips:  - Add more clothes with matching seasons and occasions.
       - Make sure shoe type matches the occasion.
       - Use neutral colors (Black, White, Grey) for more matches.

```

11. COLOR MATCHING GUIDE

The system uses the following color compatibility rules:

Color	Compatible With
Black	Everything (Neutral)
White	Everything (Neutral)
Grey / Gray	Everything (Neutral)
Navy / Dark Blue	White, Beige, Grey
Blue	Beige, Cream, Khaki
Brown	Beige, Cream, White
Green	Beige, Khaki, Cream
Maroon	Beige, White, Cream
Same Color	Always compatible with itself

Tip: To get the most outfit recommendations, add items in neutral colors (Black, White, Grey) as they match with everything.

12. OCCASION AND SEASON REFERENCE

Occasion Codes

Code	Occasion	Recommended Shoe Types
1	Wedding	Formal, Heels
2	Sports	Sports only
3	Formal	Formal, Heels

Code	Occasion	Recommended Shoe Types
4	Casual	Casual, Sandals, Sports
5	Party	Formal, Heels, Casual
6	University	Casual, Sports

Season Codes

Code	Meaning
1	Summer — Item is suitable for summer only
2	Winter — Item is suitable for winter only
3	Both — Item can be worn in summer or winter (recommended for most items)

13. LOGOUT

To log out of your account, select option 0 from the Closet Menu. You will be returned to the Main Menu and can log in with a different account or exit the program.

All your clothing items are automatically saved to the clothes.txt file every time you add a new item, so no data is lost when you log out.

14. COMMON ERRORS AND SOLUTIONS

Error Message	Solution
[ERROR] Passwords do not match!	Re-type both passwords carefully. They must be exactly the same.
[ERROR] Username already taken.	Choose a different username during signup.
[ERROR] Invalid username or password.	Check spelling. Login is case-sensitive (ahmed ≠ Ahmed).
No matching outfits found.	Add more items with matching occasions/seasons. Use neutral colors.
Your closet is empty!	Add at least 1 Upper, 1 Lower, and 1 pair of Shoes first.
[ERROR] Cannot open clothes file.	Make sure the program has write permission in the current folder.
You need at least 1 Upper, 1 Lower...	Add at least one item of each category before requesting recommendations.

15. TIPS FOR BEST RESULTS

- Add neutral-colored items (Black, White, Grey) as they match with every other color and will generate the most outfit recommendations.
- When adding items, try to mark seasonal items as 'Both' (3) where possible so they appear in more recommendations.
- Make sure your shoe type matches the occasions you assign to it. For example, do not add a Formal shoe with Occasion = Sports.

- Use consistent color names. Type 'Black' not 'Blk' or 'black'. The system uses lowercase comparison internally but consistency helps you track your closet.
- Add a variety of occasions across your clothing items so the recommendation engine has more combinations to work with.
- Each user has their own separate closet. You can create multiple accounts for different family members.

16. DATA FILES

The system stores all data in two plain text files in the same folder as the program:

users.txt

Each line contains one user account in the format:

```
username|password
```

Example:

```
ahmed|1234  
sara|abcd
```

clothes.txt

Each line contains one clothing item in the format:

```
TYPE|username|name|color|gender|season|occasion|extra_field
```

Examples:

```
UPPER|ahmed|White  
Formal Shirt|White|Boys|3|3|Cotton  
LOWER|ahmed|Black Trousers|Black|Boys|3|3|Polyester  
SHOES|ahmed|Oxford Shoes|Black|Boys|3|3|Formal
```

Do not manually edit these files while the program is running. You may corrupt your data. You can view or back them up when the program is closed.